

Report

- Assignment 3 -

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Dataset Description

Fashion-MNIST is a dataset consisting of Zalando's article images, intended as a more challenging replacement for the original MNIST dataset. It includes a training set of 60,000 examples and a test set of 10,000 examples. Each example is a 28×28 grayscale image, associated with a label from one of 10 clothing categories.

Each pixel in the image has a single integer value between 0 and 255, indicating its brightness — where higher values represent darker pixels.

The data is often provided in CSV format, where each row corresponds to one image:

- The first column contains the class label, indicating the clothing category.
- The remaining 784 columns represent the flattened pixel values ($28 \times 28 = 784$).

The class labels are as follows:

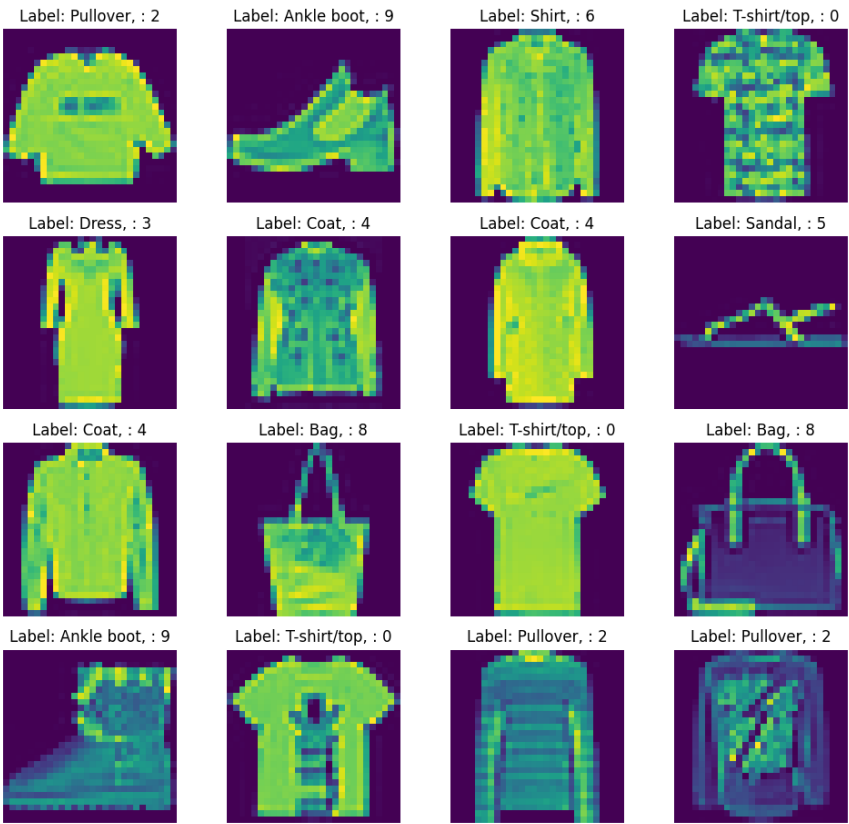
- 0: T-shirt/top
- 1: Trouser
- 2: Pullover
- 3: Dress
- 4: Coat
- 5: Sandal
- 6: Shirt
- 7: Sneaker
- 8: Bag
- 9: Ankle boot

The following is the content of the CSV file.

	label	pixel1	pixel2	pixel3	pixel4	pixel5	pixel6	pixel7	pixel8	pixel9	...	pixel775	pixel776	pixel777	pixel778	pixel779
0	2	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0
1	9	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0
2	6	0	0	0	0	0	0	0	5	0	...	0	0	0	0	30
3	0	0	0	0	1	2	0	0	0	0	...	3	0	0	0	0
4	3	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0
5 rows × 785 columns																

And, the corresponding visualization is shown below.

First 16 Images



Method for Dataset Division

I believe that distribution shift in the dataset is a critical factor that significantly affects the performance and generalization ability of machine learning models. This is because models rely heavily on the statistical properties of the training data to make predictions. If the distribution of the test data deviates from that of the training data, the model may struggle to generalize effectively, resulting in degraded performance on unseen data. Recognizing this risk, I made it a priority to examine whether such a shift exists in the dataset used for this project. To ensure consistency and fairness, I adhered to the predefined train/test split provided by Kaggle. In particular, I analyzed and compared the

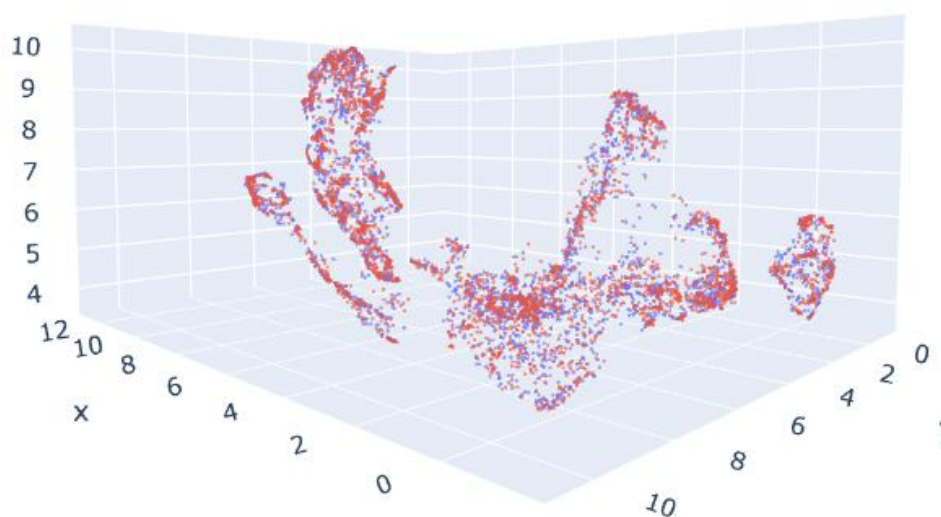
distributions of the files 'fashion-mnist_train.csv' and 'fashion-mnist_test.csv' to detect any potential discrepancies that might impact model training and evaluation.

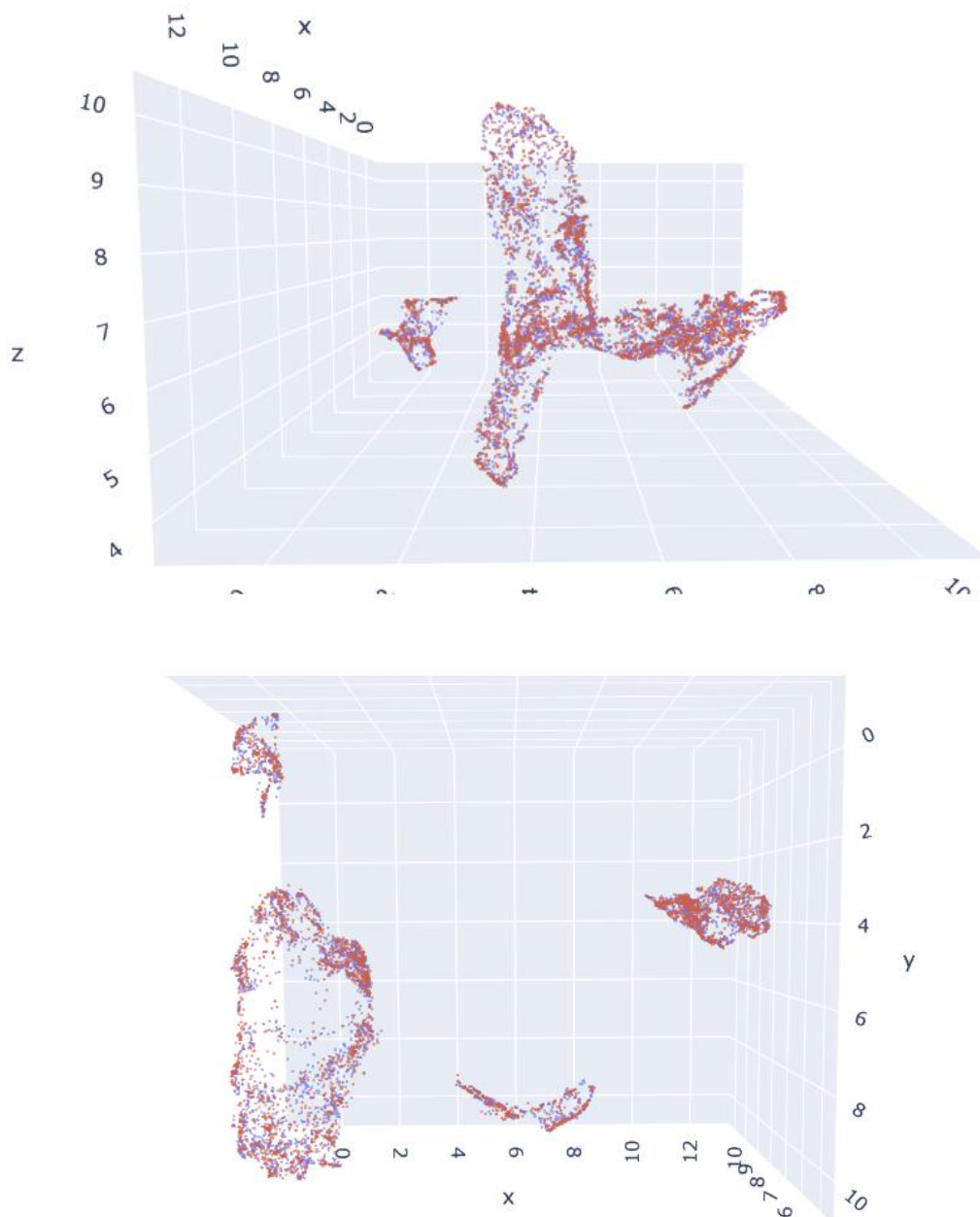
Methods for Detecting Data Distribution Shifts

To gain an intuitive understanding of potential distribution differences, I visualized the merged train and test datasets using UMAP. A 3D projection was used to allow visual inspection of the two distributions from multiple angles, making potential differences more easily observable. However, since visualizations alone may not clearly reveal distributional discrepancies, I also conducted a quantitative comparison using the Classifier Two-Sample Test (C2ST). For simplicity, logistic regression was used as the classifier with default parameters provided by the library, and accuracy was employed as the evaluation metric.

Results & Interpretation

First, the results of the UMAP visualization are shown below. Blue points represent the training data, and red points represent the test data. Since UMAP was applied in three dimensions, the resulting plot was examined from multiple angles to better capture any potential differences in distribution between the two datasets.





The distinct clusters—regardless of the point color—are presumed to correspond to different classes. Interestingly, the number of clusters is smaller than the total number of classes. This is likely because some classes with similar shapes, such as coats and t-shirts, have overlapping feature distributions, which leads to their points being grouped into the same cluster. Based on the visualization alone, there does not appear to be a significant distribution shift between the two datasets. Therefore, to quantitatively assess the difference, I performed a Classifier Two-Sample Test (C2ST). Using logistic regression as the classifier, the test yielded an accuracy of 0.8579.

Through the results of C2ST, it can be confirmed that there exists a distribution shift

between the train and test datasets, although this shift is not clearly visible through visualization. I believe this is because UMAP reduces the dimensionality of the data, which causes the information regarding the distribution differences between the two datasets to be diminished.

Furthermore, I consider the type of distribution shift between these datasets to be a covariate shift. The distribution differences arise from variations in the appearance of each piece of clothing, i.e., the covariates. Since the labels remain unchanged despite changes in the covariates, it is reasonable to classify this as a covariate shift.

Due to this distribution difference, when the validation set is created from the train set and a deep learning model is trained accordingly, the performance metric on the test set may be lower than that on the validation set.

Discussion on Monitoring & Mitigation Methods

There are various methods for monitoring distribution shifts. The visualization technique I used in this case, such as UMAP, and statistical tests like C2ST can also be employed for monitoring. In addition, feature-wise distribution comparison, which involves visualizing the distribution of each feature using histograms or density plots, can be used. Another approach is to utilize drift detection libraries that are designed to detect shifts in data distributions.

Once a distribution shift is detected through these methods, several techniques can be used for mitigation. A simple approach is to perform data rebalancing by combining all datasets, shuffling them, and then re-splitting. In other cases, techniques such as data augmentation can be applied to increase the size and diversity of the dataset, thereby helping to reduce the distribution gap.

Link for the remote source repository

<https://github.com/twoSilverUp/MLOps/tree/Assignment3>